

Silas Kwabla Gah

Department of Computer Science, University of Ghana

Legon, Accra, Ghana

gahsilas@gmail.com | +233 558 075 023 | silsgah.github.io

August 27, 2026

To: The Selection Committee and Hiring Team

Visual Computing Section, Department of Applied Mathematics and Computer Science (DTU Compute)

Technical University of Denmark (DTU)

Richard Petersens Plads, Building 324

DK-2800 Kgs. Lyngby, Denmark

Re: Application for PhD Scholarship in Computer Vision and Multimodal Human-AI Interaction (Job ID: 7907)

Dear Members of the Selection Committee,

I am writing to enthusiastically apply for the fully funded 3-year PhD scholarship in ****Computer Vision and Multimodal Human-AI Interaction**** within the Visual Computing Section at DTU Compute. With an MPhil in Computer Science (120 ECTS equivalent) and extensive doctoral research in multimodal vision-language models, 3D spatial grounding, and inference-time semantic adaptation, I am excited by DTU's mission to move beyond text-based communication and develop foundational AI systems capable of natural, two-way visual dialogue between humans and machines.

Multimodal Vision-Language Foundations & 3D Visual Grounding

My recent research focuses directly on empowering multimodal large language models (MLLMs) to interact with and ground visual environments. In recent work (*Gah et al., under submission to SACAIR 2026*), I investigated how to supervise multimodal reasoning in **Qwen2.5-VL** by deriving dense process-level feedback from frozen 2D/3D perception models (**RegionPLC**, **SAM3**, **OpenScene**). By formulating a *two-arm validate-before-train protocol*, we solved spatial specification gaming and integrated label-free geometric consistency into an online **Group Relative Policy Optimization (GRPO)** pipeline in **VERL**. This methodology of aligning generative vision-language dialogue with physical multi-view geometry is directly applicable to DTU's vision of enabling AI agents to interpret subtle visual signals, maintain multi-turn visual context, and generate grounded visual responses.

Probabilistic Calibration & Inference-Time Semantic Adaptation

In my doctoral research on **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)** (*under review at IEEE TPAMI*), I formulated a rigorous probabilistic framework to coordinate heterogeneous frozen foundation models at inference time ($\Delta\theta = 0$). By pairing Platt confidence calibration with Bernoulli uncertainty estimation and contextual reliability weighting, the framework enables vision models to dynamically decide when visual evidence should override or reinforce semantic hypotheses. This research led to training-free state-of-the-art results on open-vocabulary benchmarks (**77.9% HM** on PASCAL-5ⁱ and **33.68% HM** on ScanNet200, under review at *WACV 2026* and *ACCV 2026*). At DTU, I look forward to applying these calibration and uncertainty mechanisms to make two-way visual dialogue robust, verifiable, and transparent for human users.

Teaching Experience, Technical Rigor, and International Outlook

DTU Compute values strong scientific contributions paired with teaching and international exchange. I bring:

- **Deep Technical Proficiency:** Expert Python programmer with hands-on mastery of **PyTorch**, **CUDA**, **vLLM**, **VERL**, **Hugging Face**, **SAM/SAM3**, **DINOv2**, and modern MLLM architectures.
- **University Teaching Experience:** Over 4 years of university lecturing experience at GIMPA, where I taught undergraduate and graduate courses in **Computer Vision**, **Deep Learning**, and **Data Structures**, and supervised **55+ student research theses**.
- **Publication Track Record:** First-author manuscripts under review at premier international venues (*IEEE TPAMI*, *SACAIR/RLVR*, *WACV*, *ACCV*), demonstrating independent research capability and a commitment to publishing at top-tier conferences (CVPR, ICCV, ECCV).

I am fully prepared to relocate to Denmark, actively participate in DTU's vibrant academic community, contribute to teaching as a TA, and fulfill the international study stay component of the DTU PhD curriculum.

Thank you for your time and consideration. I welcome the opportunity to discuss how my background in multimodal perception and visual reasoning aligns with DTU Compute's research agenda.

Sincerely,

Silas Kwabla Gah